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JavaTM Technology for Palm OS® Handhelds

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Java™ technology is the platform of choice for developing end-to-end applications and deploy them on Palm OS® handhelds



- Why Java™ Technology?
- The Java 2 Platform
- MIDP for Palm OS Platform
- Java Technology in the Enterprise



- Developers are more efficient using the Java programming language than C or C++
- 30% of application developers are focusing on Java when targeting small footprint devices (Evans Data, Jan. 2001)
- Development tools available from multiple vendors, including Sun Forte for Java, Borland Jbuilder, WebGain VisualCafe, Metrowerks CodeWarrior, etc.
- Java Blueprints are best practice guides which simplify the development of scalable, available, J2EE compatible solutions.



- Java 2 Platform Enterprise Edition (J2EE™) Enables faster solution delivery time to market, eliminates vendor lock, and simplifies integration.
- J2EE is the de facto standard in the enterprise
- Best language for Web Services (Evans Data, Dec. 2001)



- There are over 30 Java-enabled handset models available from 15 manufacturers
- 15 million Java-enabled devices shipped in 2001
- Java-enabled devices to reach 721million in 2005 (Cahners MicroDesign Resources, July 2001)



Why Choose Java™ Technology?

The Java 2 Platform

MIDP for Palm OS®

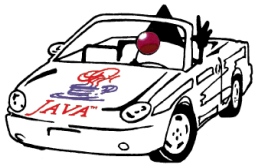
Java Technology in the Enterprise

The Java Platform



Java 2 Platform Micro Edition (J2ME™)

* Under development in JCP













Optional Packages

Java 2 Enterprise Edition (J2EE)

Optional Packages

Java 2 Standard Edition (J2SE)

Personal* Basis Profile

Personal* Profile

Foundation Profile

MIDP

CDC

CLDC

Java Card APIs

JVM

KVM

CardVM



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Java Technology and Standards



Data Services Servers
Java 2 Enterprise Edition



Communicator
High-end PDA
PersonalJava/JavaPhone



Mobile Phone
Entry Level PDA
CLDC/MIDP



SIMCard
JavaCard

3GPP MExE Technical Specifications
(Mobile Execution Environment)
Classmark 1 – WAP Environment
Classmark 2 – PersonalJava
Classmark 3 – J2ME CLDC/MIDP



- Community process to develop and evolve Java™ technology specifications
- Deliverables
  - Specification
  - Reference Implementation
  - Technology Compatibility Kit
- With or without the direct involvement of Sun

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- BEA Systems
- Cisco Systems
- Ericsson
- IBM
- Insignia
- Matsushita (Panasonic)
- Motorola
- Nokia
- Palm, Inc.
- Philips
- Research In Motion
- Siemens
- Sony
- Sun Microsystems
- Texas Instruments
- Zucotto Wireless



Why Choose Java™ Technology?

The Java 2 Platform

MIDP for Palm OS

Java Technology in the Enterprise



- Optimized Java runtime environment targeted at Palm OS handhelds
- Based on CLDC 1.0 and MIDP 1.0 specifications
- Creates a user experience comparable to Palm OS
- Performance roughly comparable to native Palm OS environment
- Released in October 2001

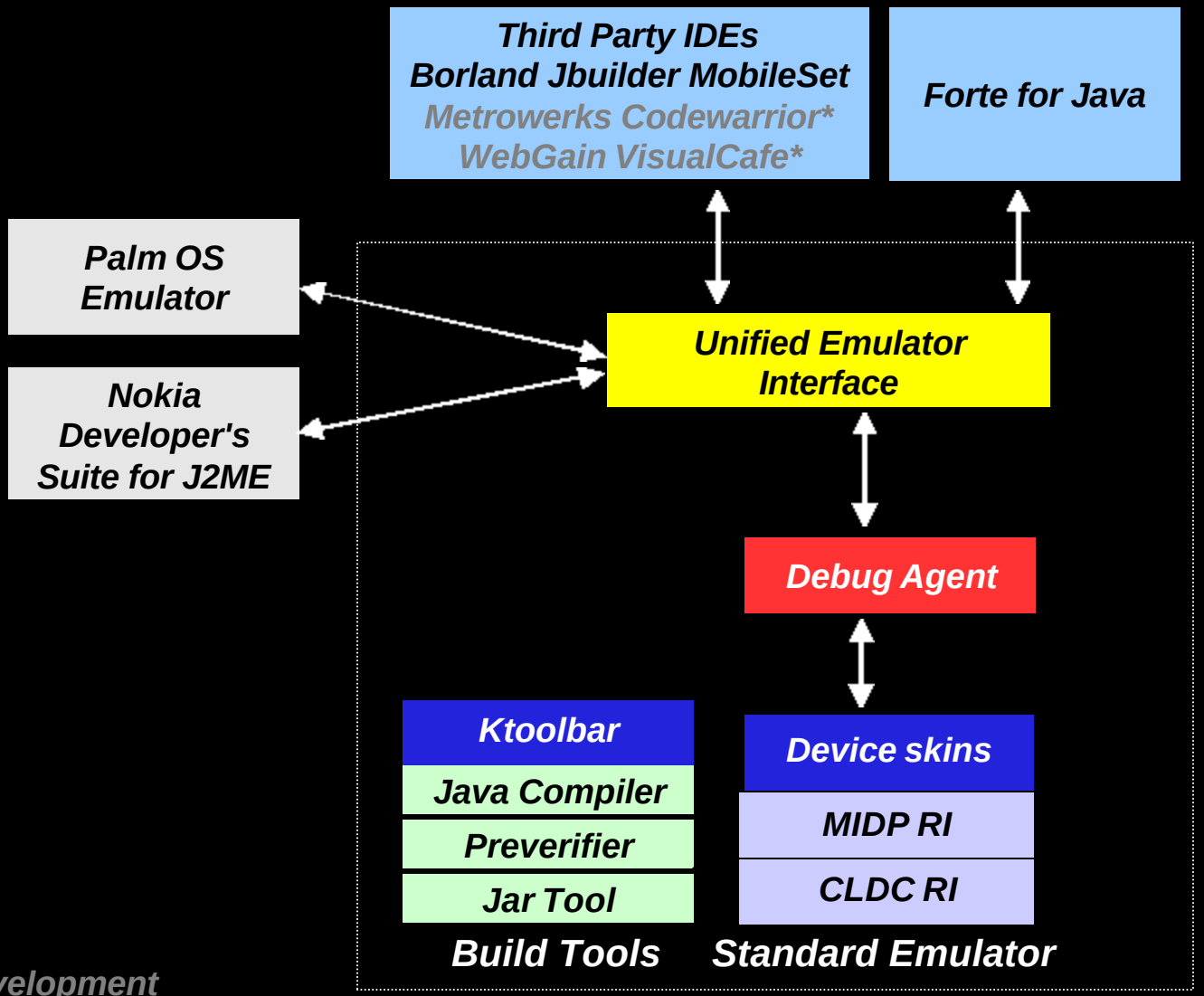
MIDP on a Phone and a Palm





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J2ME Wireless Toolkit



*In development

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- The next version of MIDP for Palm OS will include the following features
  - Over-the-Air application provisioning
  - HTTPS
  - TCP sockets and serial port support
  - Full Palm OS Version 4.0 support
  - International character set
- Early access version being demonstrated at booth #89

New APIs Standardized Through JCP

- MIDP Next Generation (JSR 118)
  - Domain security model, including signing of applications and verification of certificates
  - Network connectivity via sockets and datagrams
  - Push architecture: external events and messages routed to appropriate MIDlets
  - User Interface - extensions to low-level UI APIs to allow greater game functionality and layout control for larger screen sizes.
  - Base sound API
- Specification lead: Motorola

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New APIs Standardized Through JCP

- Wireless Telephony Communications API
  - Focusing mostly on SMS
  - Specification lead: Siemens
- Mobile Media API (JSR 135)
  - Sound and multimedia support
  - Specification lead: Nokia
- PDA Profile (JSR 75)
  - PDA specific APIs
  - Specification Lead: Palm

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Why Choose Java™ Technology?

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MIDP for Palm OS®

Java Technology in the Enterprise



Request

Provision

WAP / HTTP Server

Authentication

Profiling (personalization, etc.)

Billing

Application Management (storage, versioning, locales, etc)

Approval

Submission

J2EE-based Provisioning Server

Carrier Billing System

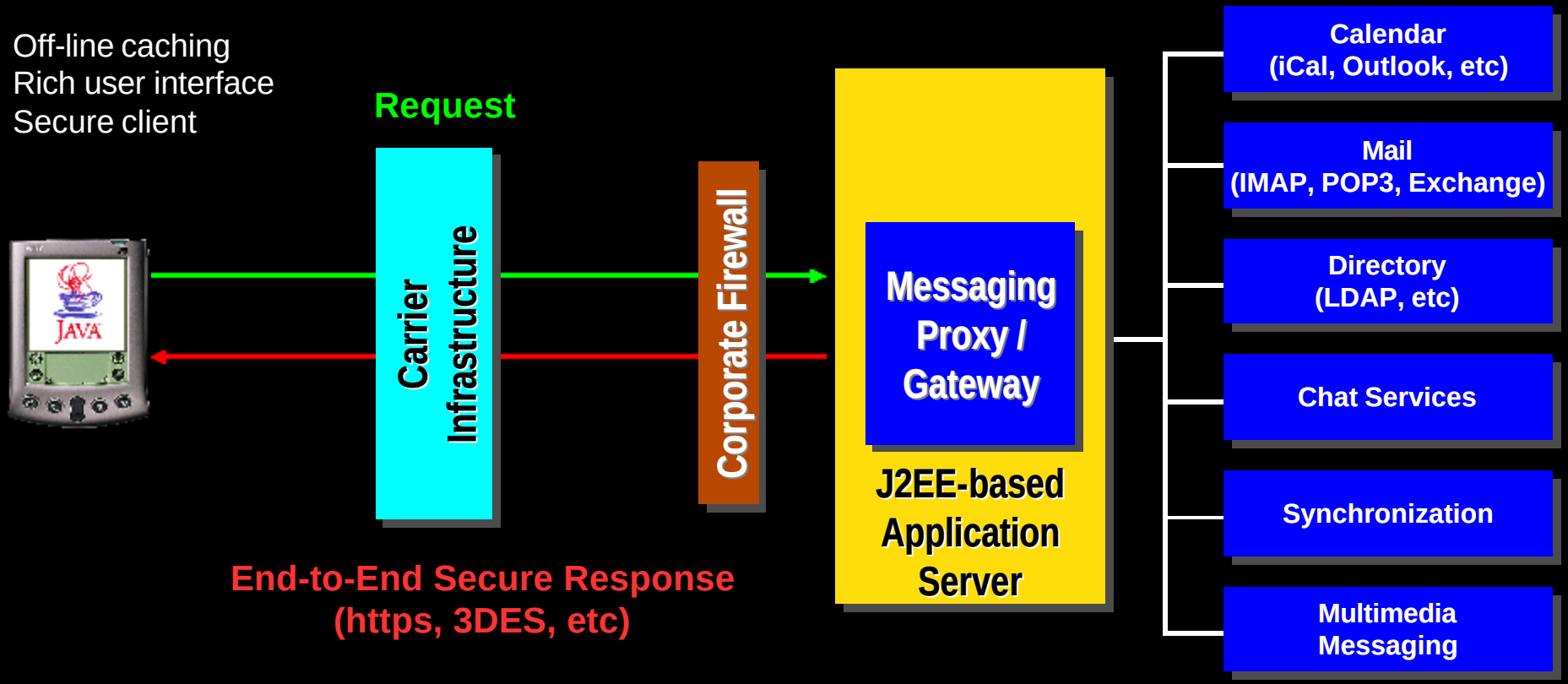
3rd Party Repository

Carrier Repository

Content Developer

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Off-line caching
Rich user interface
Secure client



Rich user interface
Local processing
Off-line browsing
Interactive client



Stand-alone or
Aggregated Content

Carrier
Portal

Carrier
Infrastructure

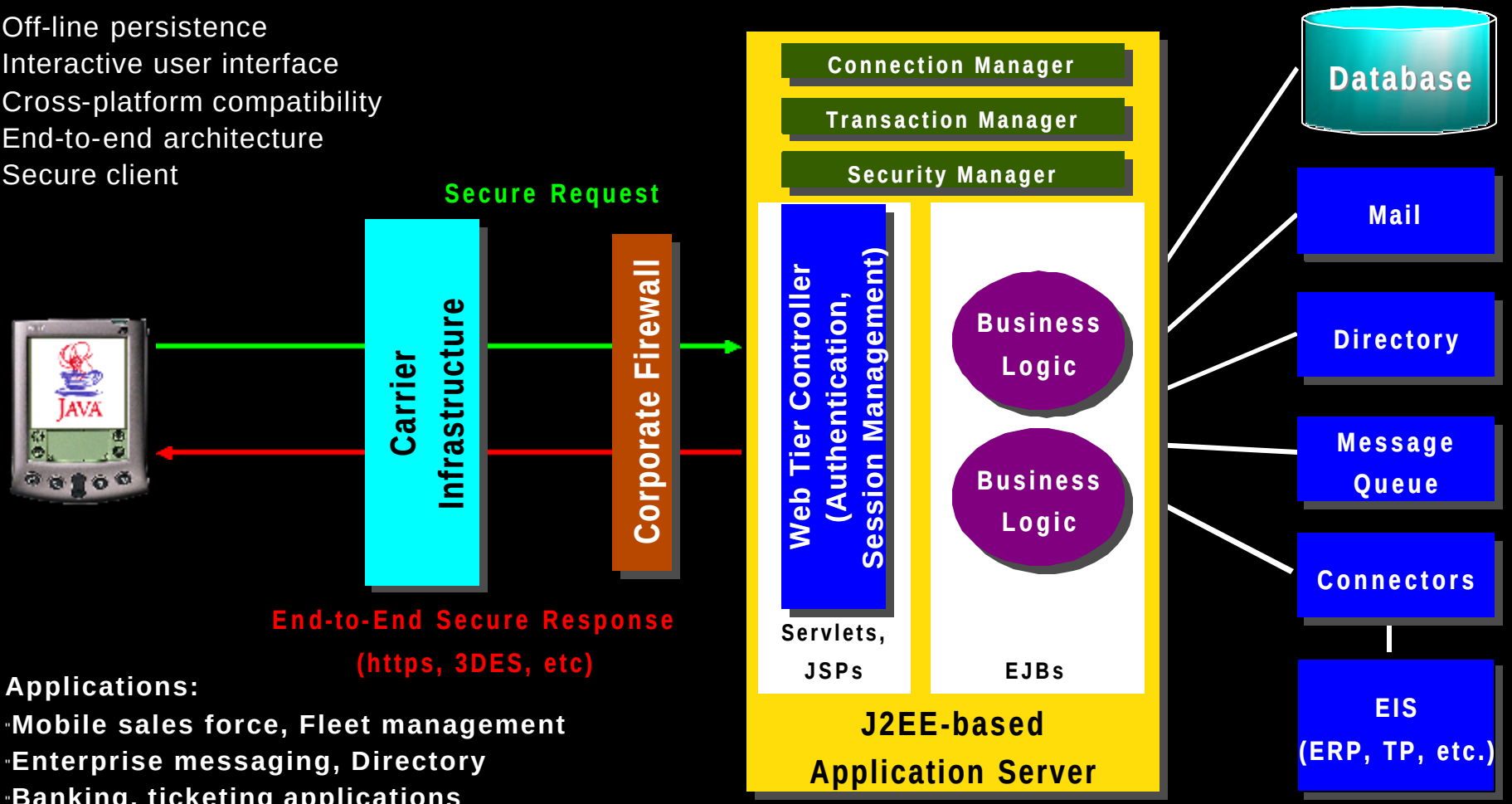
Firewall

Internet Content

Horoscope
Navigation
Stocks
Traffic
Weather
Navigation
Entertainment
News
Sports
Tourism
Shopping

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Off-line persistence
Interactive user interface
Cross-platform compatibility
End-to-end architecture
Secure client



Applications:

- **Mobile sales force, Fleet management**
- **Enterprise messaging, Directory**
- **Banking, ticketing applications**



- Gartner Group, Nov. 2000
  J2ME enabled mobile phones and PDAs will control 70% of the worldwide market by the end of 2004
- ARC group, July 2001
  J2ME VM sales will reach over 680 million units in 2004
- Cahners In-Stat/MDR, July 2001
  J2ME mobile device sales will follow a compound annual growth rate (CAGR) of 83.5% from 2000 to 2005



- Java Community Process
  http://jcp.org
- J2ME
  http://java.sun.com/j2me
- Java Wireless Developer Initiative
  http://wireless.java.sun.com



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